Could a Bigger Heart Make the Grinch Strong?

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Could a heart that is too small make someone grumpy? In How the Grinch Stole Christmas, the Grinch starts with a heart two sizes too small. When he hears Whoville singing, his heart grows three sizes that day. But would a small heart make someone grumpy? What would happen if a heart grew that fast? It depends on the reason, experts say and on whether you are a snake. Heart size is **linked** to body size, says heart doctor Michelle Kittleson. A 2025 **study** of more than 360,000 people in Australia found that people with smaller hearts were more likely to die within five years, maybe because they did not pump blood as well. Kittleson thinks the Grinch may not have exercised enough, so he felt tired and grumpy. Exercise can make a heart grow in a good way, says Leslie Leinwand. **Pregnancy** can too, but they **shrink** back within months. A heart growing in moments is more like a **python**. Leinwand studies **pythons**, whose hearts can grow up to 25 **percent** in a day after a big meal.

Word Treasure

Pregnancy -- Pregnancy is when a baby grows inside its mother before it is born.
python -- A python is a very large snake.
percent -- Percent means a part out of every hundred.
study -- A study is careful research done to learn something new.
linked -- Linked means joined or connected to something.
shrink -- Shrink means to get smaller.

Background you need

The heart is the part of the body that pumps blood around to help the body work.

A heart that is too small for its body may not pump blood as well, so a person can feel tired.

Exercise can help a heart grow stronger in a good way, and some animals, like pythons, can have hearts that grow quickly after a big meal.

Break it down

WHO: The Grinch, heart doctor Michelle Kittleson, and Leslie Leinwand, who studies pythons

WHAT: Experts explain whether a small heart could make someone grumpy and what would happen if a heart grew very fast

WHERE: Whoville, where the Grinch hears singing, and Australia, where a study of more than 360,000 people was done

WHY: Heart size is linked to body size and how well the heart pumps blood, and exercise or a big meal can make a heart grow

Why it matters

Exercise can help your heart grow stronger in a good way, so moving your body can help you feel less tired and grumpy.

Test what you remember

1. **How big was the Grinch's heart at the start of the story?**
   - (A) One size too small
   - (B) Three sizes too big
   - (C) Two sizes too small
   - (D) Just the right size
2. **What happened when the Grinch heard Whoville singing?**
   - (A) His heart shrank
   - (B) His heart grew three sizes that day
   - (C) He went to the doctor
   - (D) He ate a big meal
3. **What did a 2025 study of more than 360,000 people in Australia find?**
   - (A) People with smaller hearts were more likely to die within five years
   - (B) People with bigger hearts were always healthier
   - (C) Pythons lived longer than people
   - (D) Exercise made hearts shrink
4. **Why might people with smaller hearts be more likely to die within five years?**
   - (A) They did not pump blood as well
   - (B) Their hearts grew three sizes at once
   - (C) Their hearts were too big for their bodies
   - (D) They exercised too much
5. **What does Michelle Kittleson think made the Grinch feel tired and grumpy?**
   - (A) He ate a big meal every day
   - (B) He may not have exercised enough
   - (C) He studied pythons
   - (D) He lived in Whoville
6. **How much can a python's heart grow in a day after a big meal?**
   - (A) Two sizes
   - (B) Three sizes
   - (C) Up to 25 percent
   - (D) Up to 5 percent

Answer key (try the quiz first!): 1: C 2: B 3: A 4: A 5: B 6: C

Questions to think about

1. Michelle Kittleson: Heart size is linked to body size, and the Grinch may not have exercised enough, so he felt tired and grumpy.

2. Leslie Leinwand: Exercise can make a heart grow in a good way, and a python's heart can grow up to 25 percent in a day after a big meal.

MY THOUGHTS (at least 20 words):